

# Dongchen Wang

Postdoctoral Scholar in GPS division  
California Institute of Technology

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## Education

- 2015–2022 University of Illinois at Urbana-Champaign, USA.  
Ph.D. in **Civil and Environmental Engineering**, concentration in Water Resources Engineering and Science  
Thesis : Density-Driven Mixing Layer in Environmental Flows  
Advisor : Marcelo H. Garcia  
M.S. in **Applied Mathematics**, concentration in Computer Science and Engineering (Scientific computing track)
- 2010–2013 École Polytechnique joint with University of Nice–Sophia Antipolis, France.  
M.S. and Engineer Diploma in **Hydraulic Engineering/Hydro-informatics**
- 2006–2010 University of Science and Technology Beijing, China.  
B.S. in **Mechanical Engineering**, concentration in Thermal Energy and Dynamics Engineering

## Work Experience

- 2022–now **Postdoctoral**, *Lamb Surface Processes Group, California Institute of Technology, Pasadena, CA.*  
• NASA Delta-X project : Numerical modeling of the Mississippi River Delta's land evolution and climate risk assessment.  
• NSF/USACE project : Numerical assessment of nature-based land-building solutions at West Bay, LA.
- 2013–2014 **Engineer**, *Saint-Venant Hydraulics Lab, Electricity of France (EDF), France.*  
• Developed secondary flow correction modules in the hydro-morphodynamics HPC code OpenTelemac  
• Developed a river-meandering dynamics module in collaboration with the National Sedimentation Laboratory, USDA.  
• Simulated a river regulation project and its application to the Loire River near a nuclear power plant water intake, in collaboration with IHE–UNESCO, Delft, Netherlands.

## Research interest

My research advances fundamental theories in environmental fluid mechanics and sediment transport while providing actionable insights for water-resources and sediment management, tail-event risk mitigation, and nature-based solutions (NbS). I integrate high-performance computing models, remote sensing, and machine learning to address complex environmental challenges in both built and natural systems. I have expertise in high-performance numerical modeling of multiphase flows, sediment transport, and riverine–coastal processes using both low- and high-order solvers such as *NEK5000*, *OpenFOAM*, and *OpenTelemac*.

## Teaching Interests

Core Water Resources and Environment Engineering courses :

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|----------------------------------|----------------------------------|----------------------------------|
| – Environmental Fluid Mechanics  | – Hydraulic Analysis and Design  | – Surface Hydrology              |
| – Urban Hydrology and Hydraulics | – Pollutant Transport and Mixing | – Water and Wastewater Treatment |
| – Groundwater Modeling           | – Environmental Systems          | – Sediment Transport             |

Other foundational or advanced topics :

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|----------------------------------|-------------------------------|------------------------------|
| – Numerical Methods and Analysis | – Nonlinear Dynamical Systems | – Decision and Risk Analysis |
| – Mathematical Optimization      | – Remote Sensing              | – Machine Learning           |

## Awards & Scholarships

- Outstanding Student Presentation Award (OSPA) at the American Geophysical Union (AGU) Annual Meeting, selected by the Remote Sensing Technical Committee in the Hydrology Section, 2021 (top 2–5%)
- Best Student Paper Award at the Illinois Water Conference, 2018 (1st place)
- International Erasmus Mundus Scholarship from the European Union (EU), 2012

## In the News

- Interview by the University News Bureau on my research : Impact of Lake Michigan water level rise on complex bidirectional flow in the Chicago Area Waterway System (CAWS), 2021. *Role : Lead author.*  
*Societal impact* : Informed a potential 10% increase in the City of Chicago's allowable water use and identified a possible pathway for invasive species migration between Lake Michigan and the Mississippi River.

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## Grants

- 2023–2025 **NSF flagship supercomputer Frontera EAR23015, Rated highly competitive.** Title : The role of flocculation in building land in river deltas : Theory of microscale processes and its impact on macroscale modeling at the Mississippi Delta. PI : Michael P. Lamb. My role : Drafted grant proposal and managed resource allocation.
- 2022–2025 **NSF flagship supercomputer Frontera CTS22005, Rated highly competitive.** Title : Interfacial turbulence modeling in environmental flows. PI : Marcelo H. Garcia. My role : Drafted grant proposal and managed resource allocation.
- 2019–2023 **NSF Extreme Science and Engineering Discovery Environment (XSEDE) TG-CTS190067, Amount : US\$866,300.00,** Title : Understanding complex dynamics in engineered and natural river networks.. PI : Marcelo H. Garcia. My role : Drafted grant proposal and managed resource allocation

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## Teaching Experience

- 2019–2020 **Environmental Fluid Mechanics** (graduate level), *Role : Grader and host office hour. Instructor : Gary Parker.*

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## Teaching Training

- The Postdoc Pre-Faculty Pedagogy Preparation Program (5P) at Caltech
- Graduate Academy for College Teaching at UIUC

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## Research Projects

- Freshwater flocculation and its impact on sustaining floodplains and deltaic wetlands, in collaboration with the U.S. Army Corps of Engineers and Water Institute. (Funding source : NSF)
- Delta-X : Enabling deltas to thrive in a century of rising seas, in collaboration with NASA JPL, BU, and UT Austin. (Funding source : NASA)
- Impact of the Great Lakes watershed climate change and invasive species mitigation alternatives on the Chicago Area Waterway System, as it relates to flooding, water quality, and navigation. (Funding sources : NSF and Metropolitan Water Reclamation District of Greater Chicago (MWRD))
- Simulation of mudflow/debris flow during a 1,000-year precipitation event in the Palmdale, CA watershed : climate-related extreme weather event and vehicle safety/risk analysis. (Funding source : Private sector)
- Density-driven mixing layer and its surface pattern in the Chicago Sanitary and Ship Canal, in collaboration with the U.S. Geological Survey. (Funding sources : NSF and MWRD)
- Liquefaction of tailings dam break in Brazil and fine tailing material transport in the Paraopeba River, Brazil. (Funding source : Private sector)
- Numerical study of river channel bifurcation and sediment transport : the Bulle effect, in collaboration with Utah State University
- Impact of Large Wood Debris (LWD) on river bank stability, in collaboration with the U.S. Department of Agriculture and Louisiana State University
- Factors in turbidity current sediment deposition for large hydraulic dam design. (Funding source : Private sector)
- Tidal currents-driven pollutant and sediment transport in the Lower Passaic River, NJ. (Funding source : Private sector)

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## Journal Publications (Published)

- Zaki, A. S., Wang, D., Baker, S. R., Grotzinger, J. P., Dickson, J., & Lamb, M. P. (2025). Mars' hourglass landforms as local source-to-sink fluvial systems. *JGR Planets*. (accepted)
- Wang, D., Li, Z., Rojas-Aguirre, A. F., & García, M. H. (2021). Impact of Lake Michigan water level rise on complex bidirectional flow in the Chicago Area Waterway System (CAWS). *Journal of Great Lakes Research*, 47(6), 1626–1643.
- Dutta, S., Wang, D., Tassi, P., & García, M. H. (2017). Three-Dimensional Numerical Modeling of the Bulle-Effect : The non-linear distribution of near-bed sediment at fluvial diversions. *Earth Surface Processes and Landforms*, 42(14), 2322–2337.
- Mendoza, A., Abad, J. D., Langendoen, E. J., Wang, D., Tassi, P., & El Kadi Abderrezzak, K. E. K. (2016). Effect of sediment transport boundary conditions on the numerical modeling of bed morphodynamics. *Journal of Hydraulic Engineering*, 143(4), 04016099.
- Langendoen, E. J., Mendoza, A., Abad, J. D., Tassi, P., Wang, D., Ata, R., & Hervouet, J. M. (2016). Improved numerical modeling of river morphodynamics with steep banks. *Advances in Water Resources*, 93, 4–14.

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## Journal Publications (Under review, preprint available)

- Wang, D., Nghiem, J. A. K., Salter, G., Wright, K. A., Passalacqua, P.,... Simard, M. & Lamb, M. P. Modeling mud as flocculated and suspended bed-material load in rivers and deltas. Submitted to *JGR-Earth Surface*.
- Wang, D., Fischer, P., & Garcia, M. H. Weak buoyancy force-driven parallel mixing layer in open-channel flows. Submitted to *Journal of Fluid Mechanics*.
- Wang, D., Dutta, S., Jackson, P. R., Best, J., & Garcia, M. H. A generalized buoyancy force-driven confluence mixing model with high-resolution remote sensing image validation. Submitted to *Water Resources Research*.
- Wang, D., Tassi, P., Langendoen, E. J., Mendoza, A., Abad, J. D., Ata, R., & Hervouet, J. M. Flow and bed morphology in large-amplitude meanders : Comparison of two- and three-dimensional numerical models. Submitted to *Advances in Water Resources*.
- Noh, B., et al. Constraining the Distribution of 3D Fractal Structures in Mud Flocs. Submitted to *JGR-Earth Surface*.
- Simard, M., et al. Delta-X : An airborne remote sensing framework to evaluate land-building processes in coastal deltas. Submitted to *Remote Sensing of Environment*.
- Shukla, T., Wang, D., Rhoads, B.L. Modeling-based assessment of channel stability in an anabranching reach of a lowland meandering river. Submitted to *CATENA*.

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## Journal Publications (In preparation)

- Wang, D., Nghiem, J. A. K., Salter, G., Wright, K. A., Passalacqua, P.,... Simard, M. & Lamb, M. P. Morphodynamics model of Wax Lake Delta and its implication for coastal wetland building processes. In preparation.
- Wang, D., Esposito, C. R., Yuill, B. T., Keller, K., Foster-Martinez, M., & Lamb, M. P. A Deltaic Flow Diversion Resulted in a Decade of Net Erosion Followed by Net Accretion. In preparation.
- Wang, D., Rojas-Aguirre, A. F., & Garcia, M. H. Upscale formula for downstream-blocked depositional turbidity currents. In preparation.
- Wang, D., Waratuke, A., & Garcia, M. H. Quantifying and detecting mudflow initiation in arid watersheds : Insights from the 2015 Palmdale flash flood event. In preparation.
- Wang, D., Rojas-Aguirre, A. F., Li, Z., & Garcia, M. H. Operational Performance Graph (OPG) : A new method for composite risk assessment and optimization of hydraulic structure operations. In preparation.

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## Other Manuscripts

- Wang, D., & Garcia, M. H. (2020, August). Density-driven mixing layer in environmental flows : A high-resolution remote sensing image-based, numerical simulation and field measurements-aided confluence mixing model. In *River Flow 2020 : Proceedings of the 10th Conference on Fluvial Hydraulics* (Delft, Netherlands, 7–10 July 2020) (p. 130). CRC Press.
- Wang, D., & Garcia, M. H. (2018, August). Sediment transport and non-conservative gravity currents coupling by finite volume methods using a shock-capturing scheme. In *River Flow 2018 : Proceedings of the 9th Conference on Fluvial Hydraulics* (Lyon, France).
- Dutta, S., Tassi, P., Wang, D., & Garcia, M. (2017). Assessing the impact of the Bulle-Effect on morphodynamics of a laboratory-scale diversion using 3D numerical simulations. In *Proceedings of the XXIVth TELEMAC-MASCARET User Conference*, 17–20 October 2017, Graz University of Technology, Austria (pp. 157–157).
- Wang, D., Dutta, S., Garcia, M. H., & Jackson, P. R. (2016, January). Three-dimensional numerical modeling of mixing at the junction of the Calumet–Sag Channel and the Chicago Sanitary and Ship Canal : A comparison between density-driven and advection-driven mixing. In *International Conference on Fluvial Hydraulics, RIVER FLOW 2016* (pp. 1587–1595). CRC Press/Balkema.
- Wang, D., & Tassi, P. (2014). Secondary flow corrections for the TELEMAC-MASCARET modelling system. In *Proceedings of the 21st TELEMAC-MASCARET User Conference 2014*, 15–17 October 2014, Grenoble, France (pp. 225–233).
- Mendoza, A., Wang, D., Abad, J. D., Langendoen, E. J., Tassi, P., & El Kadi Abderrezzak, K. (2014, August). Numerical modeling of dune progression in a high-amplitude meandering channel. In *River Flow 2014* (pp. 3–5).
- Wang, D., Tassi, P., El Kadi Abderrezzak, K., Mendoza, A., Abad, J. D., & Langendoen, E. J. (2014, August). 2D and 3D numerical simulations of morphodynamic structures in large-amplitude meanders. In *River Flow 2014* (pp. 3–5).
- El Kadi Abderrezzak, K., Ata, R., Tassi, P., Wang, D., Hervouet, J. M., & Die Moran, A. (2014). 2D and 3D numerical modeling of dam-break waves over moveable beds. *11th International Conference on Hydroinformatics*.

- Mattia, F., Wang, D., Hervouet, J. M., Leopardi, A., El Kadi Abderrezzak, K., & Tassi, P. (2013). Numerical simulations of bar formation and propagation in straight and curved channels. In *XXth TELEMAC-MASCARET User Conference 2013* (pp. 85–91).

## Oral Presentations

- Wang, D. & Lamb, M. P. (2025, December). Estimation of sediment trapping efficiency of the Wax Lake Delta, Louisiana, using a high-resolution river delta morphodynamic model. In *AGU Fall Meeting* (2025, EP23B-04).
- Wang, D. (2024, February). From surface patterns to subsurface dynamics : A fluid mechanics approach to using remote sensing images to estimate density differences between two mixing flows. Caltech Geo-Club presentation.
- Wang, D., & Garcia, M. H. (2021, December). Coupled numerical modeling and remote sensing data analysis of river mixing surface patterns and sub-surface processes at the Chicago Sanitary and Ship Canal–Calumet Sag River confluence. In *AGU Fall Meeting* (2021, H42G-08).
- Wang, D., Li, Z., Rojas-Aguirre, A. F., & García, M. H. (2021, October). Impact of Lake Michigan water level rise on complex bidirectional flow in the Chicago Area Waterway System (CAWS). Water Resources Science and Engineering (WRES) Seminar Series, Department of Civil and Environmental Engineering, University of Illinois.
- Wang, D., Konsoer, K. M., Garcia, M. H., Rhoads, B. L., Langendoen, E. J., & Best, J. (2019, December). Quantifying the effects of outer-bank large wood debris (LWD) patches on flow resistance and bank erosion with a porous structure approximation. In *AGU Fall Meeting* (2019, EP51E-2155).
- Wang, D., Dutta, S., Garcia, M. H., & Jackson, P. R. (2018, June). Potential influence of density currents on index velocity ratings : A case study for the Chicago Sanitary and Ship Canal (CSSC) near Lemont, IL. 8th International Symposium on Environmental Hydraulics (ISEH).

## Poster Presentations

- Wang, D., Esposito, C. R., Yuill, B. T., Keller, K., Foster-Martinez, M., & Lamb, M. P. (2024, December). A deltaic flow diversion resulted in a decade of net erosion followed by net accretion. In *AGU Fall Meeting Abstracts* (Vol. 2024, No. 1334, pp. EP33B-1334).
- Wang, D., Nghiem, J. A. K., Salter, G., Wright, K. A., Passalacqua, P., Fagherazzi, S., ... & Lamb, M. P. (2023, December). Modeling mud as flocculated bed-material load in rivers and deltas. In *AGU Fall Meeting Abstracts* (Vol. 2023, No. 2149, pp. EP33D-2149).
- Wang, D., & Garcia, M. H. (2022, December). Turbulent mixing layer with horizontal density gradients in finite-depth channel flows. In *AGU Fall Meeting Abstracts* (Vol. 2022, pp. OS45E-1244).
- Wang, D., & Garcia, M. H. (2020, December). Density-driven confluence mixing pattern identified using high-resolution aerial images at the Chicago Sanitary and Ship Canal. In *AGU Fall Meeting* (2020, H010-0003).
- Wang, D., Konsoer, K. M., Garcia, M. H., Rhoads, B. L., Langendoen, E. J., & Best, J. (2018, December). Quantifying the effects of outer-bank large wood on flow resistance and bank erosion in an evolving meandering river. In *AGU Fall Meeting* (Vol. 2018, EP31A-04).
- Wang, D., Dutta, S., Garcia, M. H., & Jackson, P. R. (2018). Effect of density difference between the Calumet–Sag and Chicago Sanitary and Ship Canal (CSSC) on the flow structure at the diversion accounting station, Lemont, Illinois. *Illinois Water Conference*.
- Wang, D., Konsoer, K. M., Rhoads, B. L., Garcia, M. H., & Best, J. (2017, December). Numerical estimation of outer-bank resistance characteristics in an evolving meandering river. In *AGU Fall Meeting* (2017, EP41A-1821).
- Wang, D., & Garcia, M. H. (2017). Numerical modeling of the Lower Passaic River morphodynamics, New Jersey. *The Ninth International Conference on Remediation and Management of Contaminated Sediments*, Battelle.
- Wang, D., & Tassi, P. (2015). Numerical simulations of a sequence of bars in symmetric and non-symmetric large-amplitude meandering channels. In *SHF Conference : Small-Scale Morphological Evolution of Coastal, Estuarine, and River Systems*.
- Wang, D., Tassi, P., Claude, N., El Kadi Abderrezzak, K., Morphodynamics modeling and optimization of forced self bed-scour by groynes in a meandering reach (2015). In *36th IAHR World Congress*.
- Wang, D., Tassi, P., & El Kadi Abderrezzak, K. (2014, May). Numerical simulations of a sequence of bars in a large-amplitude meandering channel. In *EGU General Assembly Conference* (p. 8194).

## Professional and Academic Societies

**President (2021–2022)** of Young Professionals Network, International Association for Hydro-Environmental Engineering and Research (IAHR).

**Chair (2016–2022)** of the student chapter, International Water Resources Association (IWRA).

**Member** of ASCE, AGU, IAHR, IWRA, and AAAS.

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## Academic Services

- **Reviewer** : Earth Surface Dynamics | Journal of Hydrology | JGR – Earth Surface | Geomorphology | Water Science and Engineering
- **Department seminar coordinator (2021–2022)** for the Water Resources Engineering and Science (WRES) Seminar Series, Department of Civil and Environmental Engineering, University of Illinois
- **Co-founder (2022)** of the ILLINOIS+MINNESOTA Hydro-Geo Seminar Series—a platform for University of Illinois and University of Minnesota graduate students to discuss research topics, receive feedback from professors, exchange ideas, and practice presentation skills
- **Exhibitor and Organizer (2016–2022)** of Engineering Open House (EOH) at the University of Illinois—the largest scientific event in the Midwest that encourages the next generation, particularly underrepresented groups, to pursue careers in science and engineering (my 2021 exhibit “3, 2, 1 Lift-Off! DIY Hydraulic Pressurized Rocket” won the Best Theme Award)
- **Co-organizer** of IAHR Young Professional Network student campus visit trips. The trip to the University of Wisconsin–Madison in 2018 was featured in the IAHR News Magazine
- **Organizer (2016–2020)** of new student orientation and social events for WRES

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## Mentorship

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|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021      | Mentoring certificate training session.                                                                                                                                                                   |
| 2018–now  | Mentored graduate students and post-docs in the Hydrosystems Laboratory, UIUC, assisting with research planning, literature review, experimental equipment design, and thesis and research paper writing. |
| 2023–2025 | Mentored undergraduate students through the Summer Research Undergraduate Fellowship (SURF) program and mentored PhD students at Caltech..                                                                |